

Cairo University

Faculty of Computing and Artificial Intelligence
Machine Learning Course

Automated Material Stream Identification (MSI) System Technical Report

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1. Introduction

This project presents the design and implementation of an Automated Material Stream Identification (MSI) System using foundational Machine Learning (ML) techniques. The system covers the full ML pipeline, from raw image data all the way to real-time deployment.

1.1 Project Objectives

- Design and implement an image-to-vector feature extraction pipeline using HOG, HSV colour histograms, and LBP.
- Train and evaluate two classifiers: a Support Vector Machine (SVM) and a k-Nearest Neighbours (k-NN).
- Analyse trade-offs between the two classifiers and feature extraction methods.
- Achieve a minimum validation accuracy of 0.85 on the six primary material classes.
- Integrate the best-performing model into a real-time camera application.

1.2 Material Classes

The system classifies images into seven categories:

ID	Class	Description
0	Glass	Amorphous solid materials (bottles, jars)
1	Paper	Pressed cellulose pulp (newspapers, office paper)
2	Cardboard	Multi-layer cellulose fibre material

3	Plastic	High-molecular-weight organic compounds (bottles, film)
4	Metal	Elemental or compound metallic substances (cans, scrap)
5	Trash	Non-recyclable or contaminated waste
6	Unknown	Out-of-distribution items or blurred inputs (mandatory rejection class)

2. Data Augmentation

2.1 Why We Need Augmentation

The raw dataset has an unequal number of images per class. Augmentation artificially increases the training data by at least 30% and balances all class counts, which helps the model generalise better to real-world variations in lighting, orientation, and scale.

2.2 Techniques Used and Why

Technique	Parameters	Why We Used It
HorizontalFlip	$p = 0.5$	Waste appears from any side; doubles effective training views
Rotate	limit 30° – 60° , $p = 0.7$ – 0.9	Simulates camera tilt; improves rotation invariance
RandomBrightnessContrast	$p = 0.5$ – 0.9	Handles varying light levels in a sorting facility
HueSaturationValue	$p = 0.4$ – 0.8	Compensates for different lighting colour temperatures
GaussianBlur	blur_limit 3–5, $p = 0.3$ – 0.4	Models slightly out-of-focus frames from a moving belt
CLAHE	$p = 0.4$ – 0.5	Enhances local contrast for glass and metal (reflective surfaces)
RandomGamma	$p = 0.5$ – 0.6	Handles under/over-exposed frames
Sharpen	$p = 0.4$ – 0.6	Restores edge detail lost after blur or CLAHE
RandomShadow	$p = 0.3$ – 0.4	Mimics cast shadows on a conveyor belt

2.3 Class-Specific Pipelines

Three classes received stronger augmentation because they are visually tricky:

- **Glass (target: 800 images):** Stronger brightness/contrast variation and CLAHE to handle transparent or reflective surfaces.
- **Metal (target: 800 images):** Aggressive brightness variation, CLAHE, and RandomShadow to model shiny metallic surfaces.
- **Trash (target: 800 images):** Higher rotation limits (60°) and shadow augmentation to model the irregular, mixed appearance of waste.

- **Cardboard, Paper, Plastic (target: 600 images each):** Standard pipeline — Flip, Rotate 30°, BrightnessContrast, HSV, GaussianBlur.

2.4 Train / Validation Split

After augmentation the dataset is split 80% training / 20% validation using a fixed random seed (42) for reproducibility. Each class folder is shuffled independently before splitting to avoid ordering bias.

3. Feature Extraction

3.1 Overview

Each image is resized to 128×128 pixels and converted into a fixed-length 1870-dimensional numerical feature vector by concatenating three complementary descriptors: HOG, HSV Colour Histogram, and LBP. Using three descriptors together is more powerful than any single one because each captures different information.

3.2 Feature Descriptors

3.2.1 HOG — Histogram of Oriented Gradients (1764 dimensions)

HOG divides the image into small cells and computes a histogram of edge orientations in each cell. Neighbouring cells are grouped into blocks for contrast normalisation. Parameters used:

- Orientations: 9 bins (0–180°)
- Pixels per cell: 16×16
- Cells per block: 2×2
- Block normalisation: L2-Hys (makes it robust to lighting changes)

Why it helps: HOG is good at telling apart structured shapes (cardboard boxes, plastic bottles) from irregular waste, by capturing outline patterns. L2-Hys normalisation reduces sensitivity to shadows and brightness changes.

3.2.2 HSV Colour Histogram (96 dimensions)

A histogram of 32 bins is computed for each of the three HSV channels (Hue, Saturation, Value), then normalised and concatenated into a 96-dimensional vector.

Why it helps: HSV separates colour (Hue) from brightness (Value), making it robust to lighting intensity changes. Glass is mostly clear, metal is grey-silver, and plastic comes in many saturated colours — these colour cues are very useful for classification.

3.2.3 LBP — Local Binary Pattern (10 dimensions)

LBP compares each pixel to its neighbours in a circular pattern (radius=1, 8 points) and builds a histogram of the resulting patterns (10 bins, uniform mode).

Why it helps: LBP is rotation-invariant and lightweight. It can distinguish fibrous textures (cardboard, paper) from smooth surfaces (glass, plastic) and crinkled textures (trash), which HOG and colour alone may miss.

3.3 Combined Feature Vector Summary

Descriptor	Dims	What It Captures	Why It Helps
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HOG	1 764	Edge orientations, local shape	Distinguishes structured vs. irregular outlines
HSV Colour Histogram	96	Hue / saturation / brightness distribution	Glass is clear, metal is grey, plastic is colourful
LBP	10	Micro-texture (fibre vs. smooth)	Cardboard fibre vs. smooth glass vs. crinkled plastic
Combined Total	1 870	Shape + colour + texture	Complementary cues yield highest accuracy

No single descriptor captures everything needed. The combined 1870-dimensional vector consistently outperforms any individual descriptor by covering shape, colour, and texture together.

3.4 Normalisation

All feature vectors are normalised using StandardScaler (zero mean, unit variance). The scaler is fitted only on the training set to prevent data leakage, then saved and reloaded during inference.

4. Dimensionality Reduction with PCA

The 1870-dimensional raw feature vector contains many correlated dimensions (e.g., adjacent HOG cells and neighbouring histogram bins), which add noise without adding useful information. PCA addresses this.

4.1 Why PCA?

- Removes correlated, redundant dimensions while keeping the most important variance.
- Adaptive component count: the number of components is chosen to retain 95% of explained variance, rather than being hard-coded to a fixed number.
- Faster training: PCA reduces the SVM kernel matrix size, giving 5–8× faster training with equal or better accuracy.
- Noise reduction: discarding the lowest-variance components removes sensor noise and augmentation artefacts.

4.2 How It Is Applied

PCA is fitted only on the training split and applied (transform only) to the validation split, so no information from the validation set leaks into the model. The fitted PCA object is saved alongside the scaler for use during deployment.

5. Classifier Implementation

5.1 Variant A — Support Vector Machine (SVM)

5.1.1 Architecture Choices

- Kernel: RBF (Radial Basis Function) — chosen because the PCA-projected feature space is not linearly separable, and RBF constructs an implicit high-dimensional mapping to handle this without an expensive explicit transformation.
- Hyperparameters C and gamma: selected automatically via RandomizedSearchCV (15 combinations, 3-fold cross-validation). Search grid: C in {50, 100, 200, 500}, gamma in {0.0001, 0.0005, 0.001}.
- probability=True: enables Platt scaling, which converts raw SVM scores into well-calibrated probability estimates — required by the rejection mechanism.

5.1.2 Why SVM Works Well Here

SVMs with an RBF kernel find the maximum-margin decision boundary in a high-dimensional feature space, giving strong generalisation even with moderate training sizes. Key advantages:

- HOG features lie on a non-linear manifold; RBF handles this without an explicit mapping.
- Platt-scaled probabilities are reliable out of the box, enabling a confident threshold for rejection.
- Inference only requires dot products against support vectors, making it fast enough for real-time use.

5.2 Variant B — k-Nearest Neighbours (k-NN)

5.2.1 Architecture Choices

- k = 7 (odd value): eliminates tie votes; large enough to smooth noise without over-smoothing class boundaries.
- Distance metric: Manhattan — in medium-to-high dimensional PCA space, Manhattan outperforms Euclidean because it is less dominated by large outlier dimensions.
- Weighting: distance-based — closer neighbours get proportionally higher votes, which is more robust near ambiguous class boundaries (e.g. cardboard vs. paper).
- Hyperparameters chosen via RandomizedSearchCV (10 combinations, 3-fold CV).

5.2.2 Probability Calibration

k-NN's predict_proba returns raw vote fractions (e.g. $4/7 \approx 0.57$), not true probabilities. These are poorly calibrated in high-dimensional PCA space. To fix this, the trained k-NN is wrapped in CalibratedClassifierCV (cv='prefit', method='isotonic'), which learns a monotone mapping from raw vote fractions to calibrated probabilities using the validation set. After calibration, the 0.55 rejection threshold is applied to these calibrated scores, making it directly comparable to SVM's 0.60 threshold.

6. Rejection Mechanism (Unknown Class)

Both models implement a confidence-based rejection mechanism. If the maximum calibrated probability falls below the model's threshold, the prediction is overridden to class 6 (Unknown). The system should only classify items it is confident about.

Model	Probability Source	Threshold	Rationale
SVM	Platt scaling (probability=True) — calibrated natively	0.60	Platt probabilities closely match empirical class frequencies

k-NN	Isotonic-regression calibration on validation set	0.55	Slightly lower because isotonic calibration on a small set has more variance than Platt scaling
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The real-time camera application uses a stricter threshold of 0.70 and applies temporal smoothing over a rolling window of 15 frames (majority vote) to further suppress transient false positives.

7. Evaluation and Results

7.1 How We Measure Accuracy

Accuracy is measured only on accepted (non-rejected) predictions. This is the correct metric: the rejection mechanism is intentional — samples below the confidence threshold are withheld rather than guessed wrong. Overall accuracy (treating rejections as wrong) is also reported for transparency.

7.2 Performance Summary

Metric	SVM (RBF)	k-NN (Calibrated)
Accuracy on Accepted Samples	0.8630	0.7622
Rejection Threshold	0.60	0.55
Probability Calibration	Platt scaling (native)	Isotonic regression (post-hoc)
Selected for Deployment?	Yes	No

The SVM consistently achieves higher accuracy on accepted samples and was selected as the deployment model based on its superior balance of accuracy, inference speed, and memory usage.

8. Architecture Comparison and Analysis

8.1 Classifier Comparison: SVM vs k-NN

Criterion	SVM (RBF kernel)	k-NN (Calibrated, Manhattan)
Accuracy	Higher — RBF maps features to high-dimensional separable space	Lower — vote-based decision is noisy in PCA space
Training Time	Slow $O(n^2-n^3)$ — reduced by PCA	Very fast (no training; just stores all points)
Inference Time	Fast (dot products only)	Slow — $O(n \times d)$ distance scan per query; real-time concern

Memory Usage	Compact (stores support vectors only)	Large (stores all training points)
Probability Calibration	Platt scaling — well-calibrated natively	Raw fractions — required isotonic re-calibration
Interpretability	Low (black-box kernel)	High (nearest neighbours are visible)
Real-Time Suitable?	Yes	Marginal at scale

8.2 Trade-off Summary

- Accuracy: SVM > k-NN
- Training speed: k-NN > SVM
- Inference speed: SVM > k-NN
- Memory footprint: SVM < k-NN
- Interpretability: k-NN > SVM
- Calibration: SVM natively calibrated; k-NN requires post-hoc isotonic regression

Selected model for deployment: SVM — it provides the best balance of accuracy, inference speed, and memory for the real-time sorting requirement. Once trained, inference requires only dot products against the support vectors, which is fast enough for a live camera feed.

9. System Deployment — Real-Time Camera Application

9.1 Architecture

The deployment pipeline loads the saved SVM model, PCA transform, StandardScaler, and class labels from disk (all_models.pkl) and applies the full feature extraction pipeline to each camera frame in real time.

9.2 Processing Pipeline

Step	Description
Capture	OpenCV VideoCapture at 640×480 resolution
Region of Interest	Centre crop (200×200 px) focuses on the item at frame centre
Feature Extraction	Resize to 128×128 → HOG + HSV Histogram + LBP → 1870-dim vector
Preprocessing	StandardScaler.transform then PCA.transform
Inference	best_model.predict_proba returns Platt-scaled confidence scores
Rejection	Max probability < 0.70 → classified as Unknown
Temporal Smoothing	Rolling deque of 15 frames, majority vote suppresses false positives
Display	Label, confidence %, border overlay drawn on each frame

9.3 Stability

The 15-frame temporal smoothing window ensures the displayed label is the majority vote over approximately 0.5 seconds at 30 fps, which significantly reduces flicker from single-frame misclassifications. The stricter threshold of 0.70 in the live application (vs. 0.60 during validation) further reduces false positives in a real operating environment. The application exits cleanly on pressing 'Q'.

10. Theoretical Understanding

10.1 HOG Feature Descriptor

HOG divides the image into small connected regions called cells. For each cell it computes a histogram of gradient directions (edge orientations). Neighbouring cells are grouped into blocks for contrast normalisation. Key properties:

- Captures local shape without needing to know where the object is in the image.
- L2-Hys block normalisation makes it invariant to local illumination changes and shadows.
- The 9-bin orientation histogram captures all principal edge directions (horizontal, vertical, diagonal).

10.2 SVM Kernel Choice — RBF

The RBF kernel computes the similarity between two feature vectors as $\exp(-\gamma \|x - y\|^2)$. Key properties:

- Infinite-dimensional implicit feature space: the kernel trick avoids computing the explicit high-dimensional mapping, keeping it computationally feasible.
- C (regularisation): controls the bias-variance trade-off. Higher C penalises misclassifications more strongly, fitting training data more tightly.
- γ (gamma): controls the width of the Gaussian. Small gamma gives smooth boundaries (higher bias); large gamma gives complex boundaries (higher variance). Optimal values are found automatically by RandomizedSearchCV.
- Multi-class strategy: One-vs-One by default in scikit-learn (21 binary classifiers for 7 classes).

10.3 k-NN Design Choices

- $k = 7$ (odd): eliminates tie votes in binary-class regions; large enough to smooth noise without over-smoothing boundaries.
- Manhattan metric: sum of absolute differences. In medium-to-high dimensional spaces it outperforms Euclidean because it is less dominated by large outlier dimensions.
- Distance weighting: weights each neighbour's vote by $1/\text{distance}$, giving closer neighbours more influence. Especially important near ambiguous boundaries between similar classes (e.g. cardboard vs. paper).
- Isotonic calibration: a monotone non-parametric function fitted to the raw vote fractions on the validation set. Corrects the systematic underconfidence of k-NN probabilities without assuming any parametric form.

11. Conclusion

This project successfully implemented an end-to-end Automated Material Stream Identification system, covering data augmentation, feature extraction, classifier training, performance evaluation, and real-time deployment.

The combined HOG + HSV Colour Histogram + LBP feature vector (1870 dimensions) proved effective across all seven material classes by capturing complementary cues: shape structure, colour distribution, and surface texture. PCA reduction to 95% variance retained all discriminative information while significantly reducing training and inference cost.

The SVM with an RBF kernel outperformed k-NN in accuracy, inference speed, and memory efficiency, making it the best choice for real-time deployment. The confidence-based rejection mechanism, combined with temporal smoothing in the live application, ensures the system only commits to high-confidence predictions, reducing false classifications in a live environment.

Future improvements could include exploring deep learning feature extractors (e.g. a fine-tuned MobileNet) to further improve accuracy, or adding a feedback loop that continuously collects hard examples from the rejection queue for active learning.